



ENSAE - 3rd year, Master 2
Exam 2022-2023

RISK MEASURES

Closed book examination, no calculator - 2 hours

The exercises are independent. Each question counts for approximately 2 points. You can answer in English or in French.

Exercise 1 (General questions about the course)

To answer the following (independent) questions, recall the relevant definitions and/or give illustrative examples.

1. What is the difference between conditional and marginal risk measures? What are the advantages and disadvantages of working with conditional rather than marginal value at risk?
2. Is it desirable or even possible to have coherent risk measures for loss distributions with possibly very fat tails?
3. For a vector of financial returns, what is the advantage of replacing the assumption of Gaussian distribution by the assumption of elliptical distribution? How restrictive is the elliptical assumption?
4. Consider a stationary sequence of losses L_t associated with a value-at-risk VaR_t , for $t = 1, \dots, n$. Give the precise definition of VaR_t such that the sequence $\mathbb{1}_{\{L_t > \text{VaR}_t\}}$ for $t = 1, \dots, n$ is iid with Bernoulli distribution of parameter $\alpha \in (0, 1)$. Explain how it can be used for backtesting.
5. Let $\alpha \in (0, 1)$ and $\xi_{n,\alpha}$ the empirical α -quantile of $X_1, \dots, X_n$. Give the definitions of ξ_α and f , as well as the precise assumptions required to obtain

$$\sqrt{n}(\xi_{n,\alpha} - \xi_\alpha) \xrightarrow{\mathcal{L}} \mathcal{N}\left(0, \frac{\alpha(1-\alpha)}{f^2(\xi_\alpha)}\right)$$

as $n \rightarrow \infty$.

6. Let r be a positively homogeneous and invariant by translation risk measure and let $\mathbf{Y}$ be a random vector with an elliptical distribution on $\mathbb{R}^d$. For all real vector $\boldsymbol{\lambda} \in \mathbb{R}^d$, which relation is satisfied by $r(\boldsymbol{\lambda}'\mathbf{Y})$?

Exercise 2 (Value at risk and quantiles of returns)

Let $\mathbf{p}_t = (p_{1,t}, \dots, p_{d,t})'$ be a vector of prices of d assets at time t . The value of a portfolio of composition $\mathbf{a} = (a_1, \dots, a_d)'$ is $V_t = \mathbf{a}'\mathbf{p}_t$ at time t .

1. Let $L_{t,t+1} = V_t - V_{t+1}$ be the loss of the portfolio at horizon 1. Express $L_{t,t+1}$ as a function of the future returns $r_{i,t+1} = (p_{i,t+1} - p_{i,t})/p_{i,t}$ for $i = 1, \dots, d$, and of quantities known at time t .

We have

$$L_{t,t+1} = \sum_{i=1}^d a_i (p_{i,t} - p_{i,t+1}) = - \sum_{i=1}^d a_i p_{i,t} r_{i,t+1}.$$

2. Take a long position in the portfolio. At the risk level $\alpha \in (0, 1)$, the value-at-risk can be defined in several ways. For example, it can be defined either as : 1) a quantile $\text{VaR}_{t+1|t}(\alpha)$ of the distribution of the loss $L_{t,t+1}$, or : 2) a quantile $\widetilde{\text{VaR}}_{t+1|t}(\alpha)$ of the distribution of the portfolio log-return $r_{t+1} = \log(V_{t+1}/V_t)$.

Compare and give an interpretation for the two definitions (for simplicity, assume that the cumulative distribution functions of $L_{t,t+1}$ and r_{t+1} are continuous and strictly increasing).

By definition of the VaR, we have

$$\alpha = P(L_{t,t+1} > \text{VaR}_{t+1|t}(\alpha))$$

and

$$\alpha = P(r_{t+1} < -\widetilde{\text{VaR}}_{t+1|t}(\alpha)).$$

The two definitions are not equivalent (as shown in the next questions), but they are similar because losses and returns move in opposite directions and risk, for a long portfolio, is in the right tail of the loss distribution (i.e. in the negative tail of the return). Also note that, for a small α , both $\text{VaR}_{t+1|t}(\alpha)$ and $\widetilde{\text{VaR}}_{t+1|t}(\alpha)$ will usually be positive.

3. We now assume that, at time t , the vector $\mathbf{r}_{t+1} = (r_{1,t+1}, \dots, r_{d,t+1})'$ follows a Gaussian distribution with mean $\boldsymbol{\mu}_t$ and variance $\boldsymbol{\Sigma}_t$. Compute $\text{VaR}_{t+1|t}(\alpha)$. Discuss under which assumptions this is a conditional or marginal value-at-risk.

Using the first question and obvious notations, we have

$$L_{t,t+1} = -\mathbf{X}'_t \mathbf{r}_{t+1} \sim \mathcal{N}(\mu_t, \sigma_t^2),$$

with $\boldsymbol{\lambda}_t = (a_1 p_{1,t}, \dots, a_d p_{d,t})'$,

$$\mu_t = -\boldsymbol{\lambda}_t' \boldsymbol{\mu}_t, \quad \sigma_t^2 = \boldsymbol{\lambda}_t' \boldsymbol{\Sigma}_t \boldsymbol{\lambda}_t.$$

It follows that $\text{VaR}_{t+1|t}(\alpha) = \mu_t + \sigma_t \Phi(1 - \alpha)$. This is a conditional VaR if the distribution of $\mathbf{r}_{t+1}$ is conditional to the present and past values of the prices $\mathcal{F}_t = \sigma(p_{i,u}, u \leq t, i \in \{1, \dots, d\})$. In this case $\boldsymbol{\mu}_t \in \mathcal{F}_t$ and $\boldsymbol{\Sigma}_t \in \mathcal{F}_t$. If the distribution of $\mathbf{r}_{t+1}$, and thus $\boldsymbol{\mu}_t$ and $\boldsymbol{\Sigma}_t$, is just conditional on what is observed at time t , then the VaR is marginal.

4. Give a condition of validity of the approximation of the log-returns by the returns :

$$r_t \simeq \frac{V_{t+1} - V_t}{V_t}.$$

Then give an approximation of $\widetilde{\text{VaR}}_{t+1|t}(\alpha)$ and compare it with $\text{VaR}_{t+1|t}(\alpha)$.

If the returns are small ($\boldsymbol{\mu}_t \simeq \mathbf{0}$ and $\|\boldsymbol{\Sigma}_t\|$ small), using $\log(1 + x) = x + o(x)$, we have

$$r_t = \log \left(1 + \frac{V_{t+1} - V_t}{V_t} \right) \simeq \frac{V_{t+1} - V_t}{V_t} = -\frac{L_{t,t+1}}{V_t}.$$

We thus have the approximation

$$\widetilde{\text{VaR}}_{t+1|t}(\alpha) \simeq \frac{\text{VaR}_{t+1|t}(\alpha)}{V_t}.$$